微积分A1期中知识梳理

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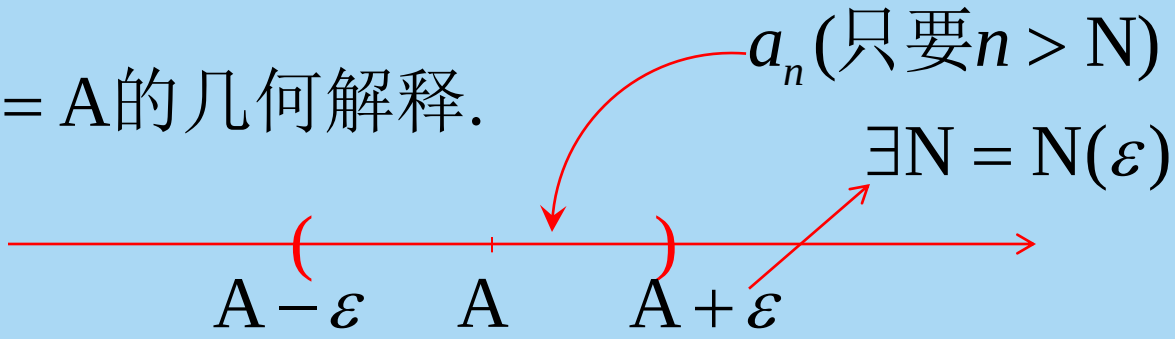
* 数列极限
* 函数极限与连续
* 导数与高阶导数
* 导数的应用

第一章 数列极限

Def. $\lim\limits_{n\to\infty} a_n = \mathrm{A}$:

$$\forall \varepsilon>0, \exists \mathrm{N}\in\mathbb{N}, \text{s.t.} \forall n>\mathrm{N}, \text{有} \left|a_n-\mathrm{A}\right|<\varepsilon.$$

Remark. $\lim\limits_{n\to\infty} a_n = \mathrm{A}$的几何解释.



$\forall \varepsilon>0, \{a_n\}$中至多有有限项落在$(\mathrm{A}-\varepsilon, \mathrm{A}+\varepsilon)$之外!

Remark. $\lim\limits_{n\to\infty} a_n = \mathrm{A}$

$$\Leftrightarrow \forall \varepsilon>0, \exists \mathrm{N} \in \mathbb{N}, \text{s.t. } \forall n > \mathrm{N}, \text{有} \left|a_n - \mathrm{A}\right| \le \varepsilon.$$

$$\Leftrightarrow \forall \varepsilon>0, \exists \mathrm{N} \in \mathbb{N}, \text{s.t. } \forall n > \mathrm{N}, \text{有} \left|a_n - \mathrm{A}\right| < \varepsilon/2.$$

$$\Leftrightarrow \forall \varepsilon \in (0,1), \exists \mathrm{N} \in \mathbb{N}, \text{s.t. } \forall n > \mathrm{N}, \text{有} \left|a_n - \mathrm{A}\right| \le 2\varepsilon.$$

$$\Leftrightarrow \forall k \in \mathbb{N}, \exists \mathrm{N} = \mathrm{N}(k) \in \mathbb{N}, \text{s.t. } \forall n > \mathrm{N}, \text{有} \left|a_n - \mathrm{A}\right| \le \frac{1}{k}.$$

Remark. ε-N语言中N的选取.

(1) 放缩法求解不等式$|a_n - \mathrm{A}| < \varepsilon$

$$|a_n - \mathrm{A}| < \cdots < \boxed{n\text{的简单表达式}} < \varepsilon$$

(2) 分段法确定N

$$\mathrm{N} = \max\{N_1, N_2, \cdots, N_k\}$$

Def. 称$\{a_n\}$发散到$+\infty$,记作$\lim\limits_{n\to\infty} a_n = +\infty$,若

$$\forall M \in \mathbb{R}, \exists N \in \mathbb{N}, 当 n > N 时, a_n > M.$$

称$\{a_n\}$发散到$-\infty$,记作$\lim\limits_{n\to\infty} a_n = -\infty$,若

$$\forall M \in \mathbb{R}, \exists N \in \mathbb{N}, 当 n > N 时, a_n < M.$$

称$\{a_n\}$发散到∞,记作$\lim\limits_{n\to\infty} a_n = \infty$,若

$$\forall M > 0, \exists N \in \mathbb{N}, 当 n > N 时, |a_n| > M.$$

Prop. 收敛列的性质

1. 收敛列的极限唯一.
2. 改变有限项,不改变数列的敛散性与极限值.
3. 收敛列的任意子列收敛到原极限
4. 收敛列必为有界列.
5. $\lim\limits_{n\to\infty} a_n = 0, \{b_n\}$为有界列,则$\lim\limits_{n\to\infty} a_n b_n = 0.$
6. 极限的保序性
7. 极限的四则运算
8. 夹挤原理

Prop.6(极限的保序性) $\lim\limits_{n\to\infty} a_n = a,\ \lim\limits_{n\to\infty} b_n = b.$

(1)若$a < b$,则$\exists N$,当$n > N$时有$a_n < b_n$.

(2)若$\exists N$,当$n > N$时有$a_n \le b_n$,则$a \le b$.

Prop.7. (极限的四则运算) 若$\{a_n\}$与$\{b_n\}$都收敛,则

(1)$\forall c \in \mathbb{R}, \lim\limits_{n\to\infty}(ca_n) = c\lim\limits_{n\to\infty} a_n$;

(2)$\lim\limits_{n\to\infty}(a_n \pm b_n) = \lim\limits_{n\to\infty} a_n \pm \lim\limits_{n\to\infty} b_n$;

(3)$\lim\limits_{n\to\infty}(a_n \cdot b_n) = \lim\limits_{n\to\infty} a_n \cdot \lim\limits_{n\to\infty} b_n$;

(4) $\lim\limits_{n\to\infty} b_n \neq 0$时, $\lim\limits_{n\to\infty}\dfrac{a_n}{b_n} = \dfrac{\lim\limits_{n\to\infty} a_n}{\lim\limits_{n\to\infty} b_n}$.

Remark. 可以推广到$\lim\limits_{n\to\infty} a_n, \lim\limits_{n\to\infty} b_n$为$\infty$或$\pm\infty$的情形, 但要右端运算有意义。

Remark. (3)$a > 0$, $a \cdot \pm\infty = \pm\infty$; $0\cdot\infty, 0\cdot\pm\infty$没有意义.

(4)$a \neq \infty, \dfrac{a}{\infty} = 0$; $a \neq 0, \dfrac{a}{0} = \infty$; $\dfrac{0}{0}, \dfrac{(\pm)\infty}{(\pm)\infty}$没有意义.

Thm.(确界原理) 非空有上界的集合必有上确界.

Thm.(单调收敛原理) 单调有界列必收敛.

Thm.(闭区间套定理) 若闭区间列$[a_n, b_n]$满足条件:

(1) $[a_{n+1}, b_{n+1}] \subset [a_n, b_n] (n = 1, 2, \cdots)$,

(2) $\lim\limits_{n\to\infty}(b_n - a_n) = 0$,

$$\boxed{\bigcap_{n\geq 1}(0, 1/n] = \phi}$$

则$\exists!\, \xi \in \mathbb{R}, s.t.\ \xi \in \bigcap\limits_{n\geq 1}[a_n, b_n]; \lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \xi.$

Thm.(Bolzano-Weirstrass定理) 有界列必有收敛子列.

Thm.(Cauchy收敛原理) 收敛列 $\Leftrightarrow$ Cauchy列.

Def.(Cauchy列)

$\{x_n\}$为Cauchy列

$$\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}, s.t. \forall n, m > N, 有 |x_n - x_m| < \varepsilon.$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}, s.t. \forall n > N, \forall p \in \mathbb{N}, 有 |x_n - x_{n+p}| < \varepsilon.$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}, s.t. \forall n \geq N, \forall p \in \mathbb{N}, 有 |x_n - x_{n+p}| \leq \varepsilon.$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}, s.t. \forall n, m > N, 有 |x_n - x_m| < 2\varepsilon$$

Remark. ε-N语言中N的选取.

$$|x_n - x_{n+p}| < \cdots < \boxed{n\text{的简单表达式, 不含}p} < \varepsilon.$$

Thm.(Stolz定理)

(1) (*/∞型)

$$\left.\begin{array}{r}\{b_n\}\text{严格}\uparrow \\ \lim\limits_{n\to\infty} b_n = +\infty \\ \lim\limits_{n\to\infty}\dfrac{a_n - a_{n-1}}{b_n - b_{n-1}} = \mathrm{A}\end{array}\right\} \Rightarrow \lim_{n\to\infty}\frac{a_n}{b_n} = \mathrm{A};$$

(2) (0/0型)

$$\left.\begin{array}{r}\{b_n\}\text{严格}\downarrow \\ \lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = 0 \\ \lim\limits_{n\to\infty}\dfrac{a_n - a_{n-1}}{b_n - b_{n-1}} = \mathrm{A}\end{array}\right\} \Rightarrow \lim_{n\to\infty}\frac{a_n}{b_n} = \mathrm{A}.$$

Remark. 常用基本结论.

(1) $\lim\limits_{n\to\infty}\sqrt[n]{n}=1$

(2) $\lim\limits_{n\to+\infty}\sqrt[n]{a}=1\,(a>0)$

(3) $\lim\limits_{n\to+\infty}\dfrac{n^b}{a^n}=0\,(a>1,b\in\mathbb{R})$

(4) $\lim\limits_{n\to\infty}\dfrac{\ln n}{n}=0,$

(5) $\lim\limits_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^n=e$

(6) $\lim\limits_{n\to+\infty}n\ln\left(1+\dfrac{1}{n}\right)=1$

(7) $\lim\limits_{n\to\infty}e^{a_n}=e^{\lim\limits_{n\to\infty}a_n}$

(8) $\lim\limits_{n\to\infty}\ln a_n=\ln(\lim\limits_{n\to\infty}a_n)$

第二章 函数极限与连续

24种函数极限的定义.

$$\lim_{x\to x_0} f(x),$$

$$\lim_{x\to x_0^+} f(x), \qquad a\in\mathbb{R},$$

$$\lim_{x\to x_0^-} f(x), \qquad +\infty$$

$$\lim_{x\to +\infty} f(x), \qquad -\infty$$

$$\lim_{x\to -\infty} f(x), \qquad \infty$$

$$\lim_{x\to \infty} f(x)$$

Def. $\lim\limits_{x\to x_0} f(x)=\mathrm{A}$: $\forall\varepsilon>0,\exists\delta\in(0,\rho),s.t.$

$$|f(x)-\mathrm{A}|<\varepsilon,\quad \forall 0<|x-x_0|<\delta.$$

$\lim\limits_{x\to x_0^-} f(x)=\mathrm{A}$: $\forall\varepsilon>0,\exists\delta\in(0,\rho),s.t.$

$$|f(x)-\mathrm{A}|<\varepsilon,\quad \forall x\in(x_0-\delta,x_0).$$

$\lim\limits_{x\to x_0^+} f(x)=\mathrm{A}$: $\forall\varepsilon>0,\exists\delta\in(0,\rho),s.t.$

$$|f(x)-\mathrm{A}|<\varepsilon,\quad \forall x\in(x_0,x_0+\delta).$$

Remark. $\lim\limits_{x\to x_0} f(x),\lim\limits_{x\to x_0^\pm} f(x)$与$f$在$x_0$的定义无关.

Def. $\lim\limits_{x\to x_0} f(x)=+\infty$: $\forall M>0, \exists \delta>0, s.t.$

$$f(x)>M, \quad \forall 0<|x-x_0|<\delta.$$

Def. $\lim\limits_{x\to\infty} f(x)=\mathrm{A}$: $\forall \varepsilon>0, \exists R>0, s.t.$

$$|f(x)-\mathrm{A}|<\varepsilon, \quad \forall |x|>R.$$

Def. $\lim\limits_{x\to-\infty} f(x)=-\infty$: $\forall M>0, \exists R>0, s.t.$

$$f(x)<-M, \quad \forall x<-R.$$

Prop. 函数极限的性质.

唯一性, 局部有界性, 保序性, 四则运算,

夹挤原理, 单调收敛原理, 复合函数的极限

Prop. （复合函数的极限）

$$\left.\begin{array}{l}\lim\limits_{x\to x_0} g(x)=u_0\\ \lim\limits_{u\to u_0} f(u)=\mathrm{A}\\ g(x)\neq u_0,\forall x\neq x_0\end{array}\right\}\Rightarrow \lim_{x\to x_0} f(g(x))=\mathrm{A}=\lim_{u\to u_0} f(u).$$

若 $\lim\limits_{u\to u_0} f(u)=f(u_0)$,则可以去掉条件“$g(x)\neq u_0,\forall x\neq x_0$”.

Thm. f在$U(x_0,\rho)$中有定义,则以下命题等价:

(1)$\forall\varepsilon>0,\exists\delta>0,\forall x,y\in U(x_0,\delta)$,有$|f(x)-f(y)|<\varepsilon$;

(2)$\exists \mathrm{A}\in\mathbb{R}$,对$U(x_0,\rho)$中任意收敛到$x_0$的点列$\{x_n\}$,有

$$\lim_{n\to\infty} f(x_n)=\mathrm{A};$$

(3) $\lim\limits_{x\to x_0} f(x)=\mathrm{A}.$

Remark. (1) $\Leftrightarrow$ (3) (函数极限的Cauchy收敛原理)

Remark. (2) $\Leftrightarrow$ (3) (用数列的极限来研究函数的极限)

Remark. $x_n \neq x_0, y_n \neq x_0, \lim\limits_{n\to\infty} x_n = \lim\limits_{n\to\infty} y_n = x_0$, 则

- $\lim\limits_{n\to\infty} f(x_n) = \mathrm{A} \neq \mathrm{B} = \lim\limits_{n\to\infty} f(y_n) \quad \Rightarrow \lim\limits_{x\to x_0} f(x)$不存在;
- $\lim\limits_{n\to\infty} f(x_n)$不存在 $\Rightarrow \lim\limits_{x\to x_0} f(x)$不存在.

- 重要不等式

$$|\sin x| \leq |x|, \forall x \in \mathbb{R}. \qquad |x| \leq |\tan x|, \forall x \in (-\frac{\pi}{2}, \frac{\pi}{2}).$$

$$|\sin x - \sin y| \leq |x - y|, \forall x, y \in \mathbb{R}.$$

$$|\cos x - \cos y| \leq |x - y|, \forall x, y \in \mathbb{R}.$$

Remark. 重要极限

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}=e,\qquad \lim_{x\to 0}(1+x)^{1/x}=e,$$

$$\lim_{x\to+\infty}\frac{\log_a x}{x^b}=0,\ \lim_{x\to 0^+}x^b\log_a x=0\ (a>1,b>0),$$

$$\lim_{x\to+\infty}\frac{x^b}{a^x}=0\ (a>1,b\in\mathbb{R}),\qquad \lim_{x\to+\infty}\frac{a^x}{x^x}=0\ (a>0,a\neq 1),$$

$$\lim_{x\to x_0}u(x)^{v(x)}=\left(\lim_{x\to x_0}u(x)\right)^{\lim\limits_{x\to x_0}v(x)}\ (\text{成立的条件?})$$

Def. 无穷大量与无穷小量 (同阶,等价,高阶)

Thm. 当$x \to 0$时:

$$(1)\ \sin x \sim \tan x \sim x;$$

$$(2)\ 1-\cos x \sim \frac{1}{2}x^2;$$

$$(3)\ \ln(1+x) \sim x;$$

$$(4)\ e^x - 1 \sim x,\ a^x - 1 \sim x\ln a\,(a>0);$$

$$(5)\ (1+x)^\alpha - 1 \sim \alpha x;$$

$$(6)\ \arcsin x \sim x;$$

$$(7)\ \arctan \sim x.$$

Remark.求函数极限的技巧

极限的四则运算

极限典式

指数-对数变换

复合函数的极限(变量替换)

等价因子替换

$o(\cdot)$的运用

夹挤原理

*Taylor*公式、洛必达法则

Def.(1)若 $\lim\limits_{x\to x_0} f(x)=f(x_0)$,则称$f$在点$x_0$处连续;

(2)若 $\lim\limits_{x\to x_0^{\pm}} f(x)=f(x_0)$,则称$f$在点$x_0$处左(右)连续.

Thm. f在点x_0处连续 $\Leftrightarrow$ f在点x_0处左连续且右连续.

Remark. f在点x_0处连续, 即$\forall\varepsilon>0,\exists\delta>0, s.t.$

$$|f(x)-f(x_0)|<\varepsilon,\quad \forall|x-x_0|<\delta.$$

Remark. f在点 x_0 处连续 $\Leftrightarrow$

对收敛到 x_0 的任意点列$\{x_n\}$,都有$\lim\limits_{n\to\infty} f(x_n)=f(x_0)$.

Def. 间断点.

$$\text{第一类间断点}\begin{cases}\text{可去间断点：} & \lim\limits_{x\to x_0} f(x)\neq f(x_0)\\ \text{跳跃间断点：} & \lim\limits_{x\to x_0^-} f(x)\neq \lim\limits_{x\to x_0^+} f(x)\end{cases}$$

第二类间断点：

$\lim\limits_{x\to x_0^-} f(x)$与$\lim\limits_{x\to x_0^+} f(x)$至少有一个不存在(或发散到$\infty$)

Thm. 有界闭区间上连续函数的性质

- 零点定理
- 介值定理
- 有界性定理
- 最大最小值定理
- 一致连续

Def.(一致连续) f在区间I上有定义,若$\forall \varepsilon > 0, \exists \delta > 0, s.t.$

$$|f(u) - f(v)| < \varepsilon, \quad \forall |u - v| < \delta, u, v \in I,$$

则称f在区间I上一致连续.

第三章 导数与高阶导数

Def. (导数, 左、右导数) 设$f(x)$在x_0的邻域中有定义.

$$f'(x_0) \triangleq \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0};$$

$$f'_-(x_0) \triangleq \lim_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0};$$

$$f'_+(x_0) \triangleq \lim_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0}.$$

Thm. f在x_0(左、右)可导$\Rightarrow$ f在x_0(左、右)连续

Remark. 奇函数的导函数是偶函数；偶函数的导函数是奇函数.

Thm. $f'(x_0)$存在 $\Leftrightarrow$ $f'_-(x_0), f'_+(x_0)$均存在且相等.

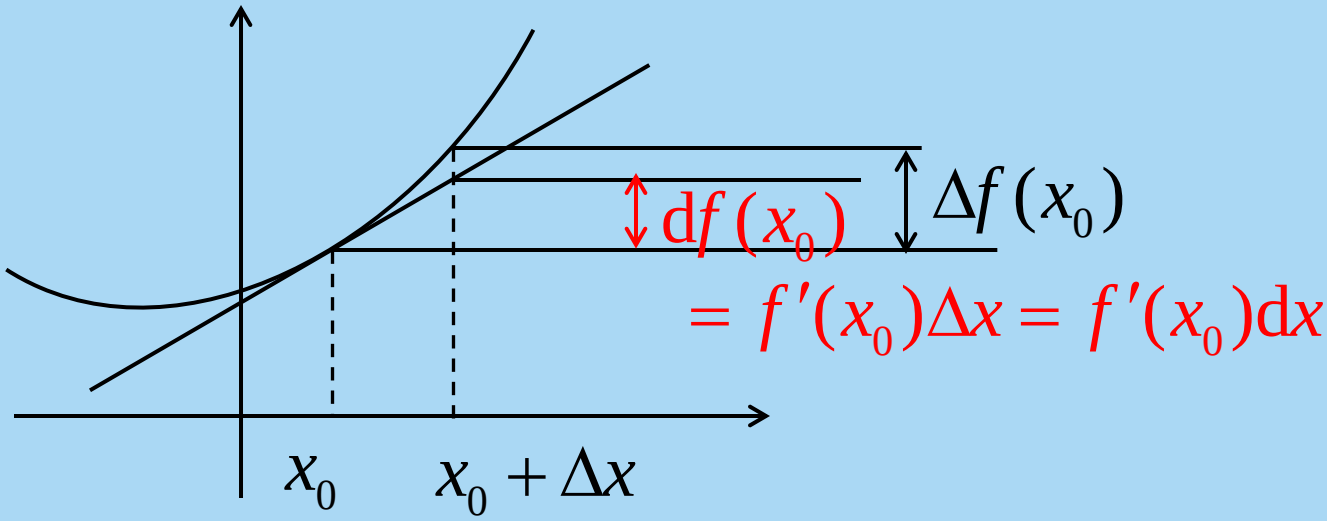
f在x_0可导时, $f'(x_0) = f'_-(x_0) = f'_+(x_0)$.

Def. 记$\Delta f(x_0) = f(x_0 + \Delta x) - f(x_0)$, 若存在常数$\alpha$, s.t.

$$\Delta f(x_0) = \alpha \Delta x + o(\Delta x) \quad (\Delta x \to 0),$$

则称f在x_0可微, 并称$\mathrm{d}f(x_0) \triangleq \alpha \Delta x \triangleq \alpha \mathrm{d}x$为$f$在点$x_0$的微分.

Thm. f在x_0可微 $\Leftrightarrow$ f在点x_0可导.



Remark. $y = f(x)$, $f'(x_0) = \dfrac{\mathrm{d}f}{\mathrm{d}x}(x_0) = \dfrac{\mathrm{d}y}{\mathrm{d}x}(x_0) = y'(x_0)$.

$$\mathrm{d}(f+g)=\mathrm{d}f+\mathrm{d}g$$

$$\mathrm{d}(cf)=c\mathrm{d}f$$

$$\mathrm{d}(fg)=g\mathrm{d}f+f\mathrm{d}g$$

$$\mathrm{d}\left(\frac{f}{g}\right)=\frac{g\mathrm{d}f-f\mathrm{d}g}{g^2}$$

Thm. f,g在x_0可导, $c\in\mathbb{R}$,则

(1)$(f+g)'(x_0)=f'(x_0)+g'(x_0)$;

(2)$(cf)'(x_0)=cf'(x_0)$;

(3)$(fg)'(x_0)=f'(x_0)g(x_0)+f(x_0)g'(x_0)$;

(4)$\left(\dfrac{f}{g}\right)'(x_0)=\dfrac{f'(x_0)g(x_0)-f(x_0)g'(x_0)}{g^2(x_0)}$.

Thm.(链式法则) $\varphi(x)$在x_0可导, $f(u)$在 $u_0=\varphi(x_0)$可导,则

$h(x)=f(\varphi(x))$在x_0可导, 且

$$h'(x_0)=f'(\varphi(x_0))\cdot\varphi'(x_0),\text{即}\ \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Remark. $f'(\varphi(x))$与$\left(f(\varphi(x))\right)'$的区别.

Thm.(反函数求导) $y(x)$存在反函数,且$y'(x)\neq 0$,则

$$x'(y)=\frac{1}{y'(x)},\frac{\mathrm{d}x}{\mathrm{d}y}=\frac{1}{\frac{\mathrm{d}y}{\mathrm{d}x}}.$$

Remark. 隐函数求导.

设$f(x,y)=0$确定了隐函数$y=y(x)$.方程

$$f(x,y(x))=0$$

两边分别对x求导.

Remark. 参数函数求导.

$$\begin{cases} x=\varphi(t) \\ y=\psi(t) \end{cases}, x=\varphi(t)\text{存在反函数,且}\varphi'(t)\neq 0,\text{则}$$

$$y'(x)=y'(t)t'(x)=\frac{y'(t)}{x'(t)}=\frac{\psi'(t)}{\varphi'(t)}. \qquad \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}}$$

Remark. 重要求导公式

$$c' = 0, \qquad (x^{\alpha})' = \alpha x^{\alpha-1},$$

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x,$$

$$(\tan x)' = \sec^2 x, \qquad (\cot x)' = -\csc^2 x,$$

$$(\sec x)' = \sec x \tan x, \qquad (\csc x)' = -\csc x \cot x$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \qquad \arctan x = \frac{1}{1+x^2}$$

$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}, \qquad \operatorname{arc}\cot x = \frac{-1}{1+x^2}$$

$$(a^x)' = a^x \ln a, \qquad (e^x)' = e^x$$

$$(\log_a x)' = \frac{1}{x \ln a}, \qquad (\ln|x|)' = \frac{1}{x}$$

$$\left(\ln\left|x + \sqrt{x^2 \pm a^2}\right|\right)' = \frac{1}{\sqrt{x^2 \pm a^2}}$$

Def. $y = f(x)$.

$$y^{(n+1)}(x) = (y^{(n)}(x))',$$

$$\frac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^{n}y}{\mathrm{d}x^{n}}\right),$$

$$f^{(n+1)}(x) = \frac{\mathrm{d}}{\mathrm{d}x}(f^{(n)}(x)).$$

Thm. 设 $f(x)$ 与 $g(x)$ 在点 x 处有 n 阶导数, $c\in\mathbb{R}$, 则

(1) $(f+g)^{(n)}(x)=f^{(n)}(x)+g^{(n)}(x)$;

(2) $(cf)^{(n)}(x)=c\cdot f^{(n)}(x)$;

(3) $(f\cdot g)^{(n)}(x)=\sum_{k=0}^{n}C_n^k f^{(k)}(x)g^{(n-k)}(x)$.(Leibniz公式)

Remark. 反函数、隐函数及参数函数求(高阶)导.

核心：复合函数求导。

$$y''(x)\neq\frac{y''(t)}{x''(t)}$$

第四章 导数的应用

§1. 微分中值定理

- (Fermat Thm) x_0是f的极值点, $f'(x_0)$存在, 则$f'(x_0)=0$.
- $f,g\in C[a,b]$, f,g在(a,b)可导, 则

(Rolle)若$f(a)=f(b)$, 则$\exists\xi\in(a,b)$, $s.t.\ f'(\xi)=0$.

(Lagrange)$\exists\xi\in(a,b)$, $s.t.\ f'(\xi)=\dfrac{f(b)-f(a)}{b-a}$.

(Cauchy)若$g'(t)\neq 0$, 则$\exists\xi\in(a,b)$, $s.t.\ \dfrac{f'(\xi)}{g'(\xi)}=\dfrac{f(b)-f(a)}{g(b)-g(a)}$.

•(Darboux Thm) f在$[a,b]$上可导, $f'_+(a) \neq f'_-(b)$,则对介于$f'_+(a)$与$f'_-(b)$之间的任意实数λ, $\exists \xi \in (a,b)$, $s.t.\ f'(\xi) = \lambda$.

Remark. $f \in C[a,b]$, f在(a,b)可导,则

(1)$\exists \xi \in (a,b)$, $s.t.$ $\quad f(b) - f(a) = f'(\xi)(b-a)$.

(2)$\forall x, x_0 \in [a,b]$, $\exists$介于x与x_0之间的ξ, $s.t.$

$$f(x) - f(x_0) = f'(\xi)(x - x_0).$$

(3)$\forall x_0, x_0 + \Delta x \in [a,b]$, $\exists \theta \in (0,1)$, $s.t.$

$$f(x_0 + \Delta x) - f(x_0) = f'(x_0 + \theta \Delta x) \cdot \Delta x.$$

Remark.构造辅助函数.

$$\left(f(x)e^{g(x)}\right)' = \left(f'(x) + f(x)g'(x)\right)e^{g(x)}.$$

§ 2. 洛必达法则

Thm. f, g在$(x_0, x_0+\rho)$中可导, $g'(x) \neq 0$, $\lim\limits_{x \to x_0+} \dfrac{f'(x)}{g'(x)} = \mathrm{A}$,则

(1)(0 / 0型) $\lim\limits_{x \to x_0+} f(x) = \lim\limits_{x \to x_0+} g(x) = 0 \Rightarrow \lim\limits_{x \to x_0+} \dfrac{f(x)}{g(x)} = \mathrm{A}.$

(2)(* / ∞型) $\lim\limits_{x \to x_0+} g(x) = \infty \Rightarrow \lim\limits_{x \to x_0+} \dfrac{f(x)}{g(x)} = \mathrm{A}.$

Thm. f, g在$(a,+\infty)$中可导, $g'(x)\neq 0$, $\lim\limits_{x\to+\infty}\dfrac{f'(x)}{g'(x)}=\mathrm{A}$,则

(1)(0/0型) $\lim\limits_{x\to+\infty} f(x)=\lim\limits_{x\to+\infty} g(x)=0 \Rightarrow \lim\limits_{x\to+\infty}\dfrac{f(x)}{g(x)}=\mathrm{A}.$

(2)(*/∞型) $\lim\limits_{x\to+\infty} g(x)=\infty \Rightarrow \lim\limits_{x\to+\infty}\dfrac{f(x)}{g(x)}=\mathrm{A}.$

Remark. 运用L'Hospital法则时注意适时分离与等价因子替换.

§ 3.Taylor公式

- $f(x)=\sum_{k=0}^{n}\frac{1}{k!}f^{(k)}(x_0)(x-x_0)^k+o\left((x-x_0)^n\right).$
- $f(x)=\sum_{k=0}^{n}\frac{1}{k!}f^{(k)}(x_0)(x-x_0)^k+\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}.$
- Taylor多项式的唯一性
- $e^x,\sin x,\cos x,\ln(1+x),(1+x)^\alpha,(1\pm x)^{-1}$的Taylor公式
- 间接展开法求Taylor公式(变量替换,待定系数,长除法)
- Taylor公式的应用.

§4.重要函数 $\sin\frac{1}{x}$

判断正误:

1. $f\in C[a,b]$, f在(a,b)中可导,则$f'_+(a)$与$f'_-(b)$都存在.

2. f在(a,b)中可导,则$f'\in C(a,b)$.

3. $f(x)=o(x^n)(x\to 0)$,则f在$x_0=0$处连续.

4. $f(x)-f(0)=o(x^n)(x\to 0)$,则$f'(0)=0$.

5. $f(x)-f(0)=o(x^n)(x\to 0)$,则$f'(0)=f''(0)=\cdots f^{(n)}(0)=0$.

只有4正确.

请扫码反馈！ 谢谢！